Expectation Propagation for Approximate Bayesian Inference

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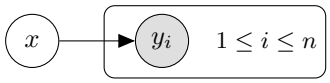
Abstract

We study Expectation Propagation (EP) and Assumed Density Filtering (ADF) for approximate Bayesian inference, following the framework introduced by Minka Minka [2013]. We extend the theoretical derivations of the clutter problem to a more general Gaussian setting and derive explicit update formulas for the algorithms. Finally, we implement EP and several competing methods and reproduce experiments with improved visualizations to compare their performances.

1 Theoretical analysis of the algorithms

1.1 Assumed-Density Filtering

The assumed-density filtering (ADF) algorithm takes in input a joint distribution $p(D, x)$, where $D = \{y_1, \dots, y_n\} \subset \mathbb{R}^d$ is observed and $x \in \mathbb{R}^d$ hidden. We assume a factorisation : $p(D, x) = p(x)\Pi_i p(y_i|x)$ corresponding to the following graphical model:



The outputs of the algorithm are the posterior over x and the evidence: $p(x|D)$ and $p(D)$. We can first think at Important sampling, but this algorithm gives the integral of any function against the desired law, but it does not give explicitly the law (with ADF we will have its density).

The algorithm: First we denote $t_0(x) := p(x)$, and $t_i(x) := p(y_i|x)$ for $i = 1, \dots, n$ such that $p(x, D) = \Pi_i t_i(x)$. Then we have to choose an approximation family of functions $\mathcal{F}$ in which $p(x|D)$ will be. For this project as we will multiply densities, so exponential families are good choices. We assume that $p(x) \in \mathcal{F}$. We will iteratively construct a density $q(x)$ approximating $p(x, D)$ in our approximation family. We initialise with $q_0(x) = p(x)$. Then we incorporate the other terms: if we have construct $q_i(x)$ for $i \in \{1, \dots, n\}$, we denote

$$\tilde{p}(x) := \frac{t_i(x)q_{i-1}(x)}{\int t_i(x)q_{i-1}(x)dx}$$

And we choose $q_i(x) \in \text{Argmin}_{q \in \mathcal{F}} KL(\tilde{p}\|q)$. Notice that for $i = 0, \dots, n-1$, we have to compute $Z_{i+1} := \int t_{i+1}(x)q_i(x)dx$, and $q_i(x) \approx \frac{t_i(x)q_{i-1}(x)}{Z_i}$, hence by induction,

$$Z_n \approx \int t_n(x) \prod_{i=1}^{n-1} \left(\frac{t_i(x)}{Z_i} \right) q_0(x)dx$$

Hence, $\Pi_i Z_i \approx \int \Pi_i t_i(x)p(x)dx = p(D)$, and we also get an approximation of $p(x|D) = p(x, D)/p(D)$.

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Example of the clutter problem: as in Minka [2013], we consider the clutter problem, but in a more general case. We did all the computations in order to gives the explicit transition steps. We consider $p(y|x) := (1-w)\mathcal{N}(y;x,I) + w\mathcal{N}(y;0,\Sigma)$ for $w \in (0,1)$, $p(x) = \mathcal{N}(x;m_0,\Sigma_0)$, and $\mathcal{F}$ is the set of all the $d-$dimensional gaussian densities.

Hence we will denote $q_i(x) = \mathcal{N}(x;m_i,\Sigma_i)$ for $i = 1,\dots,n$ and knowing the m_i and Σ_i is sufficient to solve the problem. Indeed in the gaussian setup, for any density $\tilde{p}$, the optimisation problem $\text{Argmin}_{q\in\mathcal{F}} KL(\tilde{p}\|q)$ has a solution, given by $\mathcal{N}(x;\mathbb{E}_{\tilde{p}}[X],\mathbb{E}_{\tilde{p}}[XX^T])$. Hence all the computations are trackable, we obtain the propagating steps, for all $i = 1,\dots,n$, denote $s_0 = 1$:

ADF_STEP(m_{i-1},Σ_{i-1}) :

$$\begin{array}{lcc} Z_i & = & (1-w)\mathcal{N}(y_i;m_{i-1},I+\Sigma_{i-1}) + w\mathcal{N}(y_i;0,\Sigma) \\ s_i & = & s_{i-1}\times Z_i \\ r_i & := & 1-\frac{w}{Z_i}\mathcal{N}(y_i;0,\Sigma) \\ U & := & \Sigma_{i-1}^{-1}+I \\ m_i & = & m_{i-1}+r_iU^{-1}(y_i-m_{i-1}) \\ \Sigma_i & = & \Sigma_{i-1}-r_i(\Sigma_{i-1}-U^{-1})+r_i(1-r_i)U^{-1}(y_i-m_{i-1})(y_i-m_{i-1})^TU^{-1} \end{array} \tag{1}$$

We can see that taking $\Sigma_i = v_iI$ gives the same steps as those in Minka [2013].

1.2 Expectation propagation

Expectation propagation (EP) solve the same problem as ADF, but its is more robust because it is not link with the order of the observations. ADF uses an approximation $q(x)$ in $\mathcal{F}$ of the joint, but here we will use approximation of all the factors $t_i(x)$ by some factors $\tilde{t_i}(x)$ in $\mathcal{F}$.

The algorithm: We fix $t_0(x) = p(x)$ as before, and we initialize $\tilde{t_i}(x) = 1$ for $i = 1,\dots,n$. Then we compute $q(x) := \frac{\Pi_i\tilde{t_i}(x)}{\int\Pi_i\tilde{t_i}(x)dx}$. Then until all $\tilde{t_i}(x)$ converge, we we choose a $\tilde{t_i}(x)$ to refine, then we define $q^{\setminus i}(x) := \frac{q(x)}{\tilde{t_i}(x)} \times \frac{1}{\int q(x)/\tilde{t_i}(x)dx}$ and we choose a new $q(x) \in \text{Argmin}_{q\in\mathcal{F}} KL\left(\frac{q^{\setminus i}(x)t_i(x)}{\int q^{\setminus i}(x)t_i(x)dx}\|q(x)\right)$. We denote $Z_i := \int q^{\setminus i}(x)t_i(x)dx$ and the update is: $\tilde{t_i}(x) = Z_i\frac{q(x)}{q^{\setminus i}(x)}$. And finally we recompute $q(x) = \frac{\Pi_i\tilde{t_i}(x)}{\int\Pi_i\tilde{t_i}(x)dx}$.

We the algorithm converge, we obtain the approximation $p(D) \approx \int\Pi_i\tilde{t_i}(x)dx$.

Example of the clutter problem: we assume the same hypotheses as before. We will write $\tilde{t_i}(x) = s_i\exp\left(-\frac{1}{2}(x-m_i)^T\Sigma_i^{-1}(x-m_i)\right)$. Hence the initialisation is: $\Sigma_0 = \Sigma_0, m_0 = m_0, s_0 = 1/|2\pi\Sigma_0|^{1/2}$, and for $i = 1,\dots,n$, $\Sigma_i = \infty I, m_i = 0, s_i = 1$ such that $\tilde{t_i}(x) = 1$.

So at the beginning, $q(x) = \frac{\Pi_i\tilde{t_i}(x)}{\int\Pi_i\tilde{t_i}(x)dx} = \tilde{t_0}(x) = \mathcal{N}(x;m_0,\Sigma_0)$, we initialise $m_q = m_0$, and $\Sigma_q = \Sigma_0$.

Then until all (m_i,Σ_i,s_i) converges, for $i = 1$ to n we make the steps:

$$\begin{array}{lcc} \Sigma^{\setminus i} & := & \left(\Sigma_q^{-1}-\Sigma_i^{-1}\right)^{-1} \quad \in \mathcal{S}^{++} \text{ by definition of II(§)} \\ m^{\setminus i} & := & m_q+\Sigma^{\setminus i}\Sigma_i^{-1}(m_q-m_i) \\ q^{\setminus i}(x) & = & \mathcal{N}(x;m^{\setminus i},\Sigma^{\setminus i}) \end{array}$$

Then we compute the new (m_q,Σ_q) and Z_i using (1): **ADF_STEP**$(m^{\setminus i},\Sigma^{\setminus i})$. Finally we can write the update of $\tilde{t_i}(x)$:

$$\begin{array}{lcc} \Sigma_i & = & \left(\Sigma_q^{-1}-(\Sigma^{\setminus i})^{-1}\right)^{-1} \\ m_i & := & m^{\setminus i}+(\Sigma_i+\Sigma^{\setminus i})(\Sigma^{\setminus i})^{-1}(m_q-m^{\setminus i}) \\ s_i & := & \frac{Z_i}{|2\pi\Sigma_i|^{1/2}\mathcal{N}(m_i,m^{\setminus i},\Sigma_i+\Sigma^{\setminus i})} \end{array}$$

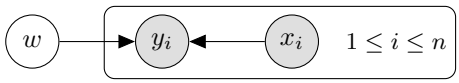
Defined like that we automatically have $q(x) \propto \prod_i\tilde{t_i}(x)$, notice that we do not know the normalizing constant, but we do not need it. At the end, when we have convergence, we can compute it. This is

our approximation of $p(D)$, given by the formula:

$$p(D) \approx |2\pi\Sigma_q|^{1/2} \exp\left(\frac{1}{2}(m_q^T \Sigma_q^{-1} m_q - \sum_i m_i^T \Sigma_i^{-1} m_i)\right) \prod_i s_i$$

Convergence of the algorithm: on the article Minka [2013], the part 3.3 explain that (EP) algorithm has fixed point, maybe non-unique, and does not always converge. We can use other ideas to find a fixed point of (EP), our we can consider a bigger approximation family $\mathcal{F}$.

Application to Bayes point matching: one very interesting application of (EP) presented in Minka [2013] is Bayes Point Machine (BPM), it consists in solving the linear classification problem: find the best $w \in \mathbb{R}^d$ such that $y = \text{sign}(w^T x)$, with a dataset $D = \{(x_1, y_1), \ldots, (x_n, y_n)\}$. The associate graphical model is now:



We assume $p(w) = \mathcal{N}(w; 0, I)$, and $p(y_i|x_i, w) = \phi(y_i w^T x_i/\epsilon)$ where $\phi(z) = \int_{-\infty}^{z} \mathcal{N}(z; 0, 1)dz$. Indeed it corresponds to the model $y = \text{sign}(w^T x + \tilde{\epsilon})$ where $\tilde{\epsilon} \sim \mathcal{N}(0, \epsilon^2)$ (we allow some mistake on the data). We are in the setup of (EP), hence we have an algorithm to find an approximation of the posterior $p(w|D)$ in a class of densities $\mathcal{F}$. The idea of (BMP) is to take the average classifier: $\text{sign}(\mathbb{E}_{w|D}[w]^T x)$ to classify some new data point x.

2 Experiments

In this section we present numerical experiments illustrating the performance of EP algorithm. Our goal was primarily to reproduce the main experiments from Minka [2013], while implementing the algorithms ourselves and providing clearer visualisations of the results. In particular, we focused on reproducing the clutter problem experiments and the Bayes Point Machine experiment described in the original paper.

All experiments were implemented in Python using `NumPy` and `Matplotlib`. Compared to the original results, our implementation improves the readability of the figures by using modern plotting tools and clearer graphical layouts.

The main quantities of interest in the clutter problem are the evidence $p(D)$ and the posterior mean $\mathbb{E}[x|D]$. For comparison we implemented the same competing methods considered in Minka [2013]: Laplace approximation, variational Bayes (VB), importance sampling, and Gibbs sampling. Accuracy is measured as the absolute difference from the true evidence or posterior mean. In Minka [2013], computational cost is measured in terms of floating point operations (FLOPs). Since Python does not provide a direct and reliable way to count FLOPs for these implementations, we instead use the number of iterations as a proxy for computational cost.

2.1 Clutter problem: comparison with other inference methods

Our first experiment reproduces the standard clutter problem experiment. The likelihood is given by

$$p(y|x) = (1 - w)\mathcal{N}(y; x, I) + w\mathcal{N}(y; 0, \Sigma),$$

with Gaussian prior $p(x) = \mathcal{N}(x; 0, 100I)$. The objective is to estimate the posterior mean $\mathbb{E}[x|D]$ and the model evidence $p(D)$.

We compare EP with four other approximate inference techniques: (1) Laplace approximation, (2) Variational Bayes, (3) importance sampling (using the prior as proposal), and (4) Gibbs sampling with latent clutter variables.

Figure 1 presents the results of a typical run of the experiment. The figure contains four subplots comparing accuracy versus computational cost for both the posterior mean estimation and the evidence estimation, for two dataset sizes ($n = 20$ and $n = 200$). As expected, the deterministic methods

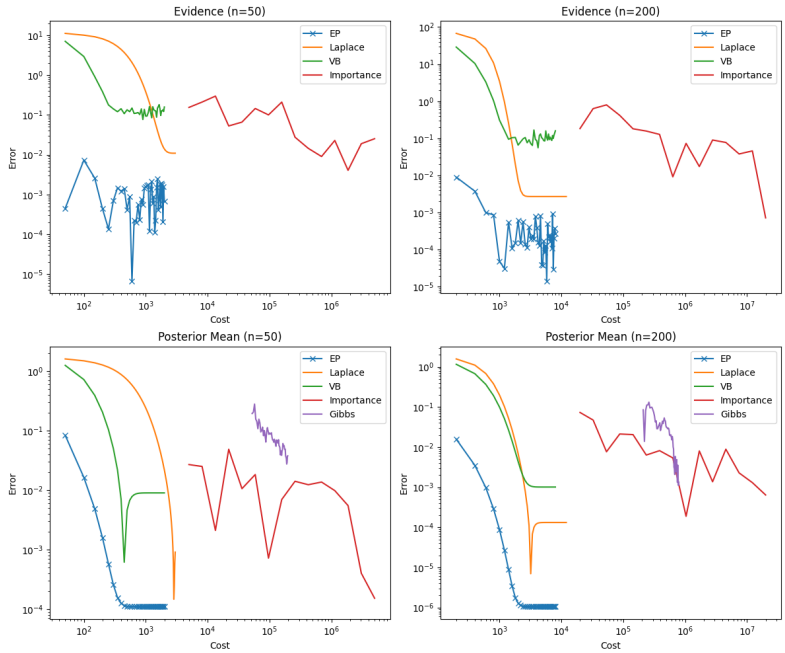


Figure 1: Comparison of approximate inference methods on the clutter problem.

(EP, Laplace, VB) benefit significantly from larger datasets, since the posterior becomes closer to Gaussian as n increases. Sampling-based approaches make fewer assumptions about the posterior but cannot exploit this increasing Gaussian structure.

Our results closely match those reported in Minka [2013], while the plots provide improved clarity thanks to modern visualization tools.

2.2 Atypical run with multimodal posterior

The second experiment studies a setting where the posterior distribution is multimodal. We consider a dataset with $n = 20$ observations for which the true posterior exhibits two distinct modes. In this regime, Gaussian approximations become problematic because they are inherently unimodal. As a result, deterministic algorithms such as EP, Laplace approximation, and variational Bayes may converge to an incorrect approximation that captures only a single mode of the posterior.

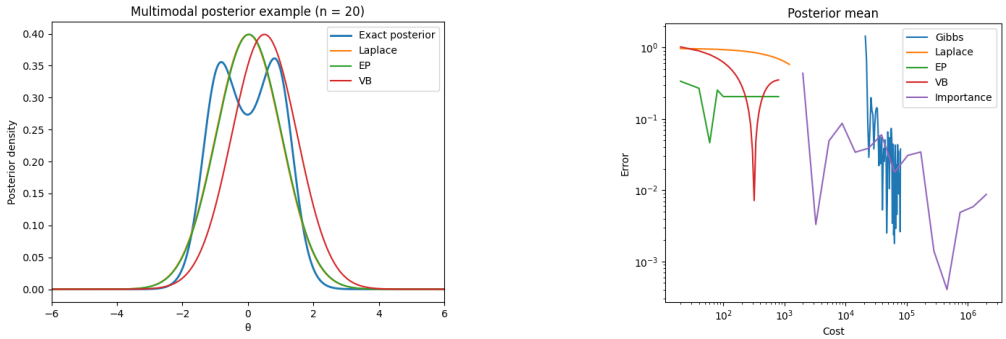
Figure 2 illustrates such an atypical run. The exact posterior shows two clearly separated modes, while the Gaussian approximations collapse onto one of them. In our experiments, the standard EP algorithm occasionally fails to converge in this regime, consistent with the behavior described in Minka [2013]. This experiment highlights a general limitation of deterministic approximate inference methods: when the posterior is highly multimodal, Gaussian approximations can fail to capture important structure in the distribution.

2.3 Bayes Point Machine experiment

Our final experiment applies EP to the Bayes Point Machine (BPM) model for linear classification.

Figure 3a shows a simple example with three training points. In addition to the two plotted coordinates, each point includes a constant third coordinate equal to 1, allowing the classifier to learn a bias term. The Bayes point classifier corresponds to an average over all possible linear separators weighted by their posterior probability.

Figure 3b compares the computational cost and error of several algorithms used to estimate the Bayes point: (1) Expectation Propagation (EP), (2) the billiard algorithm, (3) the TAP algorithm, and (4) the mean-field (MF) approximation.

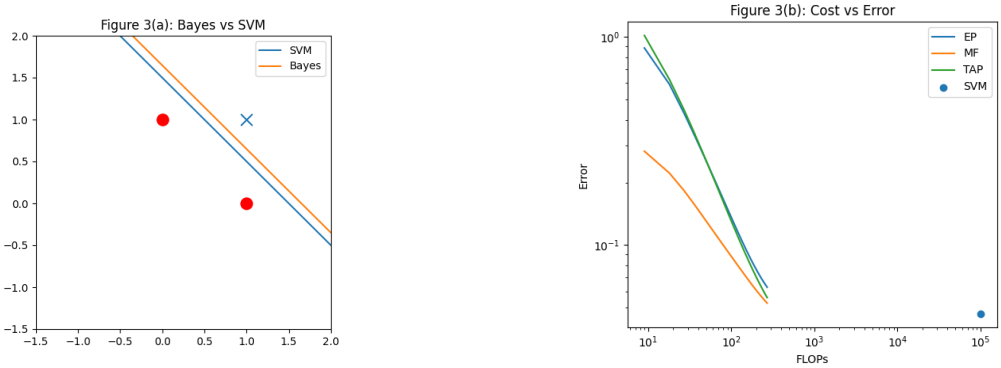


(a) Posterior comparison

(b) Cost vs accuracy

Figure 2: Example of a case with bimodal posterior distribution.

The error is measured as the Euclidean distance to the exact solution obtained by importance sampling. For reference we also plot the error corresponding to the SVM solution. Our results confirm the behaviour reported in Minka [2013]: EP converges rapidly and achieves high accuracy, while TAP and MF converge more slowly. As expected, EP and TAP converge to essentially the same solution.



(a) Posterior comparison

(b) Cost vs accuracy

Figure 3: Example of the Bayes Point Machine classification problem with three training points.

Overall, these experiments demonstrate that EP provides an efficient and accurate method for approximate Bayesian inference, while our implementation reproduces the qualitative behaviour described in the original work and improves the visual clarity of the results.

3 Conclusion

In this project we studied the ADF and EP algorithms for approximate Bayesian inference and extended the theoretical analysis of the clutter problem to a broader class of models. Our implementation reproduces the behavior reported in the original work and confirms that EP provides an efficient and accurate approximation compared with several alternative methods. Overall, the experiments highlight both the practical strengths of EP and the limitations of Gaussian approximations in multimodal settings.

References

Thomas P. Minka. Expectation propagation for approximate bayesian inference, 2013. URL `https://arxiv.org/abs/1301.2294`.